

Satyabrata Das

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Education

University of Florida

M.S., Artificial Intelligence Systems

Aug 2025 – May 2027

Gainesville, FL

- Coursework: Machine Learning for AI, Applied Deep Learning, AI Systems, AI Safety & Robustness, Computer Vision.

Experience

Evy — Computer Vision Engineer, Eye Tracking & ML

May 2026 – Present

- Built real-time eye tracking and blink detection systems for AI interview monitoring using Python, OpenCV, and MediaPipe; improved gaze estimation accuracy and reduced false positives across varied lighting, webcams, and user environments.
- Prototyped PyTorch and TensorFlow models that classified user behavior from gaze and blink data and implemented anomaly detection, enabling the system to flag suspicious interview patterns with higher reliability.
- Optimized PyTorch inference pipelines to reduce latency, integrated browser-based computer vision using WebGL, and worked with founders to launch the solution into a deployment supporting multiple concurrent users.

SouthEnd Psychiatry — IT & Operations Contractor

Sep 2024 – Aug 2025

- Provided IT support and administered the EHR and practice-management systems using Python and SQL, handling patient intake, scheduling, and processing for a telehealth group while maintaining strict confidentiality, which ensured compliance with privacy regulations and streamlined workflow.

ARC Document Solutions — Software Engineer

Jul 2022 – Sep 2024

- Engineered real-time WebSocket sync layer for ARC Facilities (20K+ MAU), cutting collaboration latency by 25% and reducing support tickets by ~30% quarter-over-quarter.
- Optimized ARC Print by integrating 5 payment gateways (Razorpay, Paytm, Cashfree, Braintree, Elavon) and reCAPTCHA, reducing failed transactions by 15%.
- Applied multithreading and Core Data in iOS apps, using Git for version control, unit testing, and CI/CD pipelines, which improved app responsiveness and lowered crash reports.

SCI-BI Software Pvt Ltd — Software Trainee

Jul 2021 – Dec 2021

- Developed ETL and automated data-processing pipelines handling millions of records daily with 99%+ accuracy.
- Optimized SQL workflows and configured Bash scripts with Prometheus and Grafana monitoring, improving data availability and reducing processing latency for large-scale datasets.

Shipped Projects

SummarIQ — NLP Pipeline for Document Intelligence

GitHub

- Built end-to-end NLP pipeline using HuggingFace Transformers (T5/BART) and LangChain for automated classification and semantic retrieval over 200+ documents, improving pipeline accuracy by ~18%.
- Deployed via FastAPI at sub-1.4s latency with FAISS-based chunking and embedding strategies for scalable semantic search and modular CI/CD-ready architecture.

Intelligent Bistro — AI Restaurant Ordering App

GitHub

- Built a full-stack app (React Native/Expo + TypeScript, Node.js/Express) with an AI assistant that converts natural-language orders into structured JSON cart actions using the Google Gemini API.
- Added intelligent item/quantity parsing and a rule-based fallback parser so the app degrades gracefully when the LLM fails, with Zustand state management.

JuggleIQ — CV Soccer Juggling Coach

GitHub

- Built a full-stack app where users upload a video and a YOLO + MediaPipe + Kalman-filter pipeline detects the ball, tracks pose, counts left/right foot touches, and returns skill metrics and coaching tips.
- Built a FastAPI (GPU) backend and a React + Vite + Recharts dashboard visualizing skill score, rhythm, and touch timeline, with annotated-video replay.

Technical Skills

- **AI & ML:** PyTorch, TensorFlow Lite, HuggingFace Transformers, Scikit-learn, LLMs, NLP, CUDA, RAG, vector databases.
- **Computer Vision:** OpenCV, MediaPipe, YOLO, 3D Pose Estimation, Eye Tracking, Gaze Estimation, Video Analytics.
- **Programming & Tools:** Python, C++, Swift, SQL, FastAPI, Docker, REST APIs, CI/CD, Git, Linux, Prometheus/Grafana.